Numerical Optimization
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Exercise sheet 9 for Numerical Optimization

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Chapter 1

Solutions Exercise sheet 9

1.1 Exercise 1

1. **Part (a): Objective and constraint functions in the form of (1).**
   Let $x = (x_1, x_2)^\top \in \mathbb{R}^2$ and choose $D = \mathbb{R}^2$. The objective function is

   $$f(x) = x_1 + x_2. \tag{1.1}$$

   The inequality constraint $x_1^2 \le 1 - x_2^2$ is equivalent to $x_1^2 + x_2^2 - 1 \le 0$, hence define

   $$g_1(x) = x_1^2 + x_2^2 - 1. \tag{1.2}$$

   The equality constraint $x_1 = x_2$ can be written as $x_1 - x_2 = 0$, hence define

   $$h_1(x) = x_1 - x_2. \tag{1.3}$$

   Therefore, the index sets are

   $$\mathcal{I} = \{1\}, \qquad \mathcal{E} = \{1\}. \tag{1.4}$$

2. **Part (b): Sets satisfying each constraint and the feasible set F.**
   The set satisfying the first constraint $x_1^2 \le 1 - x_2^2$ is

   $$S_1 = \left\{(x_1, x_2) \in \mathbb{R}^2 \,\middle|\, x_1^2 + x_2^2 \le 1\right\}, \tag{1.5}$$

   i.e., the closed unit disk centered at the origin.
   The set satisfying the second constraint $x_1 = x_2$ is

   $$S_2 = \left\{(x_1, x_2) \in \mathbb{R}^2 \,\middle|\, x_1 = x_2\right\}, \tag{1.6}$$

   i.e., the straight line through the origin with slope 1.
   The feasible set is the intersection

   $$F = S_1 \cap S_2 = \left\{(x_1, x_2) \in \mathbb{R}^2 \,\middle|\, x_1^2 + x_2^2 \le 1,\ x_1 = x_2\right\}, \tag{1.7}$$

   which is the line segment of $x_1 = x_2$ inside the unit disk.

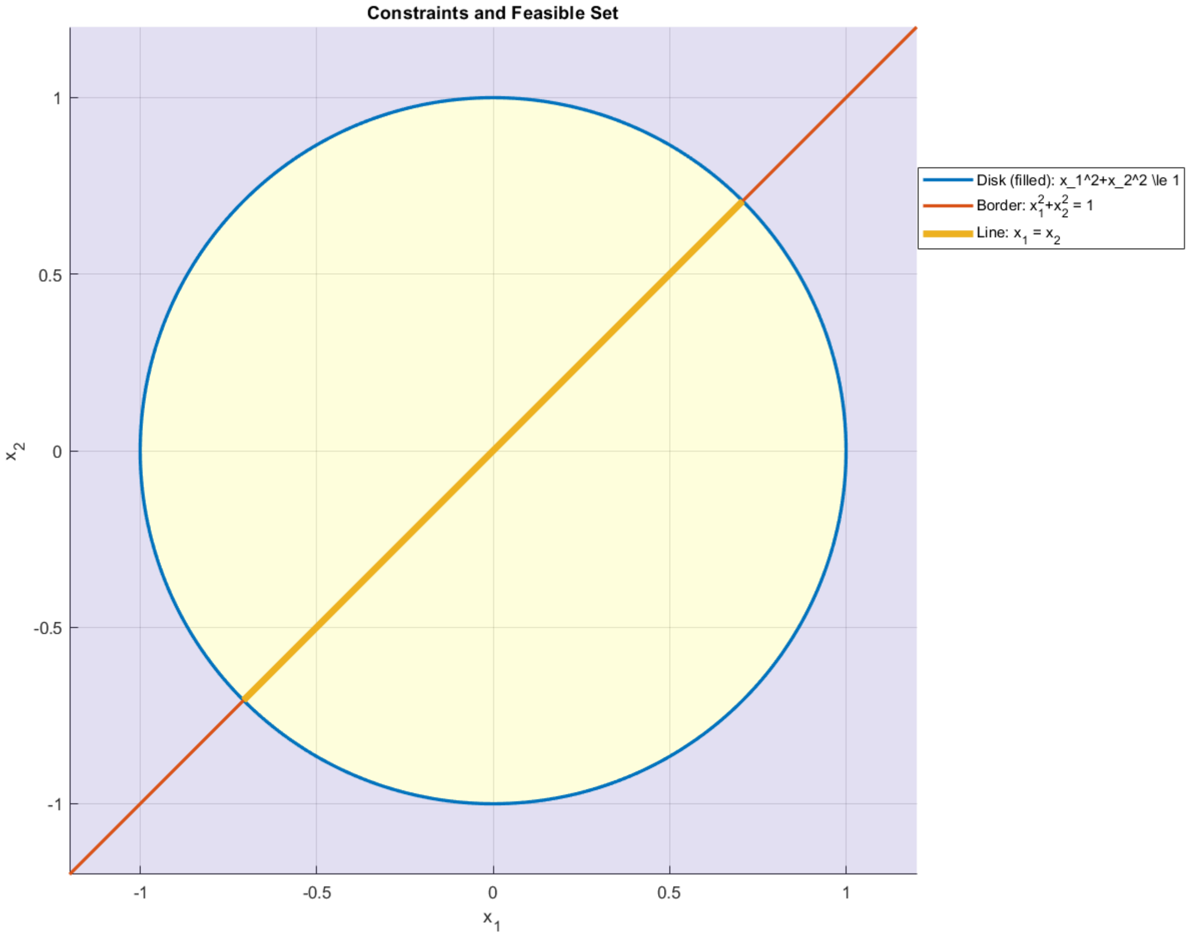

Figure 1.1: Intersection

3. **Part (c): Unique minimizer and corresponding objective value.**

   On F we have $x_2 = x_1$, so the objective becomes

$$f(x) = x_1 + x_2 = 2x_1. \tag{1.8}$$

   The feasibility condition $x_1^2 + x_2^2 \leq 1$ with $x_2 = x_1$ gives

$$2x_1^2 \leq 1 \quad \iff \quad -\frac{1}{\sqrt{2}} \leq x_1 \leq \frac{1}{\sqrt{2}}. \tag{1.9}$$

   Minimizing $2x_1$ over this interval yields the smallest value at the left endpoint,

$$x^\star = \left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right). \tag{1.10}$$

   The corresponding optimal objective value is

$$f(x^\star) = -\sqrt{2}. \tag{1.11}$$

1.2 Exercise 2

1.2.1 Summary: The Core Idea of KKT Conditions

The main goal of this exercise sheet is to understand how to find the lowest point (minimum) of a function when you are not free to move everywhere. This is called constrained optimization. Think of finding the lowest point in a valley, but you cannot cross a river or jump over a fence. These are your constraints.
The KKT conditions are a set of rules that act as a checklist. If you have found the best possible point (the local minimum) within the allowed area, that point **must** satisfy these conditions.
Here is the step-by-step logic your professor wants you to learn:

1. **Where can I go? The Cone of Feasible Directions (C_{fd})**
   - **Simple Idea:** If you are standing at a point x_0 inside the allowed region (the feasible set $\mathcal{F}$), which directions can you take a tiny step in without immediately leaving the region?
   - **Analogy:** You're standing on the edge of a park. The set of all directions you can step in that keep you inside the park is the *cone of feasible directions*.
2. **Where do I want to go? The Cone of Descent Directions (C_{dd})**
   - **Simple Idea:** From your current point x_0, which directions go "downhill"?
   - **Analogy:** You're on a hillside. The set of all directions that take you to a lower altitude is the *cone of descent directions*. The gradient (∇f) points uphill, so any direction away from the gradient is a descent direction.
3. **The First Optimality Condition (and why it's not enough)**
   - **The Golden Rule:** If you are at the lowest possible point (x^*), there cannot be a direction that is **both feasible (allowed) and a descent direction (downhill)**.
   - **In Math:** $C_{dd}(x^*) \cap C_{fd}(x^*) = \emptyset$. (The intersection is empty).
   - **The Problem:** In real life, calculating the exact shape of the feasible cone (C_{fd}) is extremely difficult for complex boundaries. It's not a practical tool.
4. **The Practical Solution: Linearizing the Problem (The Linearizing Cone, C_l)**
   - **Simple Idea:** Since the real boundary is too complex, let's approximate it with straight lines (tangents) at our point x_0.
   - **Analogy:** Instead of worrying about the complex, curvy edge of the park, you just draw a straight line where you are standing and pretend that's the boundary. It's much simpler.
   - This new, simplified cone of "allowed" directions is the *linearizing cone*. It's easy to calculate using the gradients of the constraints.
5. **The Final Step: The KKT Conditions**
   - **The Magic Connection:** The condition using the simple linearizing cone ($C_{dd}(x^*) \cap C_l(g, h, x^*) = \emptyset$) is mathematically equivalent to the famous **KKT Conditions**.
   - **What are KKT conditions in simple words?** Imagine you are at the solution point x^*, pushed against one or more "walls" (constraints). At this exact point:
     (a) **Balance of Forces:** The force pushing you downhill (the gradient of your function, ∇f) must be perfectly balanced by the forces from the walls you are touching (the gradients of the active constraints, ∇g_i). The numbers λ_i (Lagrange multipliers) tell you *how much* each wall is pushing back.

(b) **Complementary Slackness:** A wall can only push you if you are actually touching it. If you are not touching a wall (the constraint is inactive, $g_i(x) < 0$), its pushing force (λ_i) must be zero.
(c) **One-Way Force:** The walls (inequality constraints like $g_i(x) \le 0$) can only *push* you out, they can't *pull* you in. This means their force multipliers must be positive ($\lambda_i \ge 0$).

In short for me: "you can't go downhill anymore" is hard to use directly. So, we approximate the problem with lines, which magically turns into a system of simple equations (the KKT conditions). These equations are the practical tool we use to find candidate points for the solution.

1.2.2 Part a)

Big picture . We study a constrained optimization problem of the form

$$\min_{x \in \mathbb{R}^n} f(x) \quad \text{subject to} \quad g_i(x) \le 0 \ (i = 1, \dots, m), \qquad h_j(x) = 0 \ (j = 1, \dots, p), \tag{1.12}$$

with feasible set

$$\mathcal{F} \ = \ \{x \in \mathbb{R}^n : \ g_i(x) \le 0 \ \forall i, \ \ h_j(x) = 0 \ \forall j\}. \tag{1.13}$$

In unconstrained optimization, we may move in any direction. Under constraints, only certain directions keep us inside $\mathcal{F}$. The exercise builds the chain

$$\text{feasible directions} \ \Rightarrow \ \text{descent directions} \ \Rightarrow \ \text{a necessary condition for a local minimizer} \tag{1.14}$$

$$\Rightarrow \ \text{linearization cone} \ \Rightarrow \ \text{KKT conditions.} \tag{1.15}$$

1.2.3 Part (a) Show that $C_{fd}(x_0)$ is a cone

Proof. Let $d \in C_{fd}(x_0)$. By definition, there exists $\delta > 0$ such that

$$x_0 + td \in \mathcal{F} \qquad \forall t \in [0, \delta]. \tag{1.16}$$

Let $\alpha > 0$ and define $\tilde{d} := \alpha d$. Choose $\tilde{\delta} := \delta/\alpha > 0$. For any $t \in [0, \tilde{\delta}]$ we have $\alpha t \in [0, \delta]$, hence

$$x_0 + t\tilde{d} \ = \ x_0 + t(\alpha d) \ = \ x_0 + (\alpha t)\, d \ \in \ \mathcal{F}. \tag{1.17}$$

Therefore $\tilde{d} \in C_{fd}(x_0)$ and $C_{fd}(x_0)$ is a cone.

Intuition. If you can move a little bit along d without leaving $\mathcal{F}$, then you can also move along αd, you just need a smaller step size.

1.2.4 Part (b) Active constraint and its influence on C_{fd}

Key rule

- If the inequality is inactive at x_0 (i.e. $g(x_0) < 0$), then locally you are inside the feasible set and

$$C_{fd}(x_0) = \mathbb{R}^n. \tag{1.18}$$

- If the inequality is active at x_0 (i.e. $g(x_0) = 0$), then $C_{fd}(x_0)$ is typically restricted to directions that do not point outside the set. For smooth g, a common first-order restriction is

$$C_{fd}(x_0) \subseteq \{d \in \mathbb{R}^n : \ \nabla g(x_0)^\top d \le 0\}. \tag{1.19}$$

At a point that lies strictly inside the feasible region, the inequality must be inactive and the feasible directions look like all directions. At a point on the boundary, the inequality is active and the feasible directions look like a "half-space" or a "fan" of directions that stay in the region. (Here you decide activeness at $p_0 = (0, 0)$ and $p_1 = (\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$ using the specific inequality from Exercise 1 and your feasible-set sketch.)

Definition 2 (Cone of descent directions) The cone of descent directions at x_0 is

$$C_{dd}(x_0) \ = \ \{ d \in \mathbb{R}^n : \ \nabla f(x_0)^\top d < 0 \}. \tag{1.20}$$

It contains directions along which f decreases to first order.

1.2.5 Part (c) If x^* is a local minimizer, then $C_{dd}(x^*) \cap C_{fd}(x^*) = \emptyset$

Proof (contradiction). Assume x^* is a local minimizer of (1), but suppose there exists

$$d \in C_{dd}(x^*) \cap C_{fd}(x^*). \tag{1.21}$$

Since $d \in C_{fd}(x^*)$, there exists $\delta > 0$ such that $x^* + td \in \mathcal{F}$ for all $t \in [0, \delta]$. Since $d \in C_{dd}(x^*)$, we have $\nabla f(x^*)^\top d < 0$. By differentiability of f, the directional derivative condition implies that for sufficiently small $t > 0$,

$$f(x^* + td) < f(x^*). \tag{1.22}$$

But $x^* + td$ is feasible, so this contradicts that x^* is a local minimizer over $\mathcal{F}$. Hence the intersection must be empty:

$$C_{dd}(x^*) \cap C_{fd}(x^*) = \emptyset. \tag{1.23}$$

1.2.6 Part (d) Meaning of (c) in simple words, and why it is not sufficient

Explain (c) geometrically and conceptually, using the plotted points in Figure 1 as intuition. Then explain why the condition is necessary but not sufficient.

Simple meaning. At a constrained local minimizer x^*, *every direction you are allowed to move* (feasible direction) fails to decrease the objective (is not a descent direction). In other words, "there is no feasible way to go downhill."

Why it is not sufficient. The condition

$$C_{dd}(x^*) \cap C_{fd}(x^*) = \emptyset \tag{1.24}$$

is only a necessary condition. It can hold even when x^* is not a true local minimizer, because:

- The geometry of $\mathcal{F}$ can be non-smooth or complicated, making feasible directions subtle.
- Even in smooth settings, local first-order conditions can miss second-order effects.
- Practically, $C_{fd}(x^*)$ is typically hard to compute exactly, so the condition is not directly usable.

Definition 3 (Linearizing cone)

For $x_0 \in \mathcal{F}$, define the linearizing cone by

$$C_\ell(g, h, x_0) \ = \ \Big\{ d \in \mathbb{R}^n : \ \nabla g_i(x_0)^\top d \leq 0 \, \forall i \in A(x_0), \ \nabla h_j(x_0)^\top d = 0 \, \forall j \Big\}. \tag{1.25}$$

1.2.7 Part (e) What is the linearizing cone in simple words?

Explain what $C_\ell(g, h, x_0)$ means intuitively and relate it to the pictures in Figure 2.

Intuition. $C_\ell(g,h,x_0)$ is the set of directions that satisfy the constraints to *first order*, i.e. after linearizing the constraints at x_0:

- For an active inequality $g_i(x) \le 0$, the linearization is $\nabla g_i(x_0)^\top d \le 0$, meaning you do not move outward across the boundary to first order.
- For an equality $h_j(x) = 0$, the linearization is $\nabla h_j(x_0)^\top d = 0$, meaning you move tangent to the equality manifold.

In Figure 2, when x_0 is on a smooth boundary, C_ℓ typically looks like a half-space. At a corner where multiple inequalities are active, C_ℓ is the intersection of multiple half-spaces, forming a "fan-shaped" cone.

1.2.8 Part (f) Show that $C_{fd}(x_0) \subseteq C_\ell(g,h,x_0)$

Prove that every feasible direction automatically satisfies the first-order (linearized) constraint conditions.

Proof sketch using one-dimensional functions. Let $d \in C_{fd}(x_0)$. Then there exists $\delta > 0$ such that $x_0 + td \in \mathcal{F}$ for all $t \in [0,\delta]$.
Active inequalities. Fix $i \in A(x_0)$, so $g_i(x_0) = 0$ and $g_i(x_0 + td) \le 0$ for $t \in [0,\delta]$. Define $\phi(t) := g_i(x_0 + td)$. Then $\phi(0) = 0$ and $\phi(t) \le 0$ for small $t \ge 0$. Hence the right-derivative at 0 must satisfy $\phi'(0) \le 0$. By the chain rule,

$$\phi'(0) = \nabla g_i(x_0)^\top d, \tag{1.26}$$

therefore $\nabla g_i(x_0)^\top d \le 0$.
Equalities. Fix any j. Since $h_j(x_0 + td) = 0$ for $t \in [0,\delta]$, define $\psi(t) := h_j(x_0 + td)$. Then $\psi(t) \equiv 0$ on $[0,\delta]$, so $\psi'(0) = 0$. Again by the chain rule,

$$\psi'(0) = \nabla h_j(x_0)^\top d = 0. \tag{1.27}$$

Thus $d \in C_\ell(g,h,x_0)$, proving

$$C_{fd}(x_0) \subseteq C_\ell(g,h,x_0). \tag{1.28}$$

1.2.9 Part (g) Two concrete constrained problems

We are given

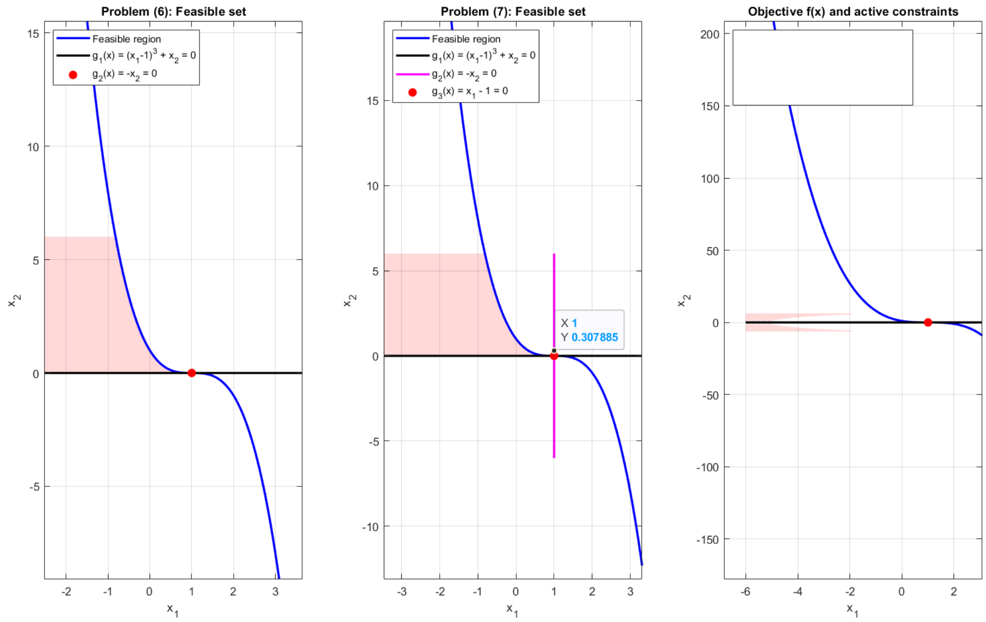

Figure 1.2: Intersection

$$f(x) = (x_1 - 2)^2 + x_2^2, \qquad g_1(x) = (x_1 - 1)^3 + x_2, \qquad g_2(x) = -x_2, \qquad g_3(x) = x_1 - 1. \quad (1.29)$$

Consider the problems

$$\min f(x) \ \text{ subject to } \ g_1(x) \le 0, \ g_2(x) \le 0, \tag{1.30}$$

and

$$\min f(x) \ \text{ subject to } \ g_1(x) \le 0, \ g_2(x) \le 0, \ g_3(x) \le 0. \tag{1.31}$$

(g-i) Draw the feasible sets The constraints rewrite as:

- $g_2(x) \le 0 \iff -x_2 \le 0 \iff x_2 \ge 0.$
- $g_1(x) \le 0 \iff (x_1 - 1)^3 + x_2 \le 0 \iff x_2 \le -(x_1 - 1)^3.$

Therefore the feasible set for problem (6) is

$$\mathcal{F}_6 = \Big\{(x_1, x_2) : \ x_2 \ge 0, \ \ x_2 \le -(x_1 - 1)^3\Big\}. \tag{1.32}$$

Since $x_2 \ge 0$ and $x_2 \le -(x_1 - 1)^3$ must be compatible, we need $-(x_1 - 1)^3 \ge 0$, i.e.

$$x_1 \le 1. \tag{1.33}$$

So one can equivalently describe

$$\mathcal{F}_6 = \Big\{(x_1, x_2) : \ x_1 \le 1, \ \ 0 \le x_2 \le -(x_1 - 1)^3\Big\}. \tag{1.34}$$

For problem (7), the additional constraint is $g_3(x) \le 0 \iff x_1 - 1 \le 0 \iff x_1 \le 1$, which is already implied by feasibility of (6). Hence $\mathcal{F}_7 = \mathcal{F}_6$ as sets, but the *active set bookkeeping* changes.

(g-ii) Why is the minimizer unique and what is it? Note that $f(x)$ is the squared Euclidean distance to $(2,0)$. Within $\mathcal{F}_6$, we can always choose $x_2 = 0$ (which minimizes x_2^2) and still remain feasible if $x_1 \leq 1$. Thus the problem reduces to minimizing

$$\min_{x_1 \leq 1} (x_1 - 2)^2, \tag{1.35}$$

whose unique minimizer is $x_1 = 1$. Therefore the unique minimizer for both (6) and (7) is

$$x^* = (1, 0). \tag{1.36}$$

(g-iii) Cone of descent directions at x^* Compute the gradient:

$$\nabla f(x) = \big(2(x_1 - 2),\, 2x_2\big), \qquad \Rightarrow \qquad \nabla f(1,0) = (-2, 0). \tag{1.37}$$

Hence

$$C_{dd}(x^*) = \{d : \nabla f(x^*)^\top d < 0\} = \{d : (-2,0) \cdot (d_1, d_2) < 0\} = \{d :\ -2d_1 < 0\} = \{d :\ d_1 > 0\}. \tag{1.38}$$

So descent means "moving to the right".

(g-iv) Cone of feasible directions at x^* We analyze feasibility of a small step $x^* + td = (1,0) + t(d_1, d_2)$ for $t > 0$.
From $x_2 \geq 0$ we get

$$0 + td_2 \geq 0 \qquad \Rightarrow \qquad d_2 \geq 0. \tag{1.39}$$

From $g_1(x) \leq 0$, we require

$$g_1(x^* + td) = ((1 + td_1) - 1)^3 + (0 + td_2) \leq 0, \tag{1.40}$$

i.e.

$$(td_1)^3 + td_2 \leq 0 \qquad \Longleftrightarrow \qquad d_2 + t^2 d_1^3 \leq 0. \tag{1.41}$$

Letting $t \to 0^+$ forces $d_2 \leq 0$. Combining with $d_2 \geq 0$ yields $d_2 = 0$. Then the inequality becomes $t^3 d_1^3 \leq 0$ for all small $t > 0$, hence $d_1 \leq 0$. Therefore

$$C_{fd}(x^*) = \{(d_1, 0) :\ d_1 \leq 0\}. \tag{1.42}$$

So feasible motion is "to the left along the x_1-axis".

(g-v) Verify condition (3) at x^* From (g-iii) and (g-iv),

$$C_{dd}(x^*) = \{d :\ d_1 > 0\}, \qquad C_{fd}(x^*) = \{(d_1, 0) :\ d_1 \leq 0\}. \tag{1.43}$$

Hence

$$C_{dd}(x^*) \cap C_{fd}(x^*) = \emptyset, \tag{1.44}$$

which matches the necessary condition for a local minimizer.

(g-vi) Linearizing cones for (6) and (7) Compute gradients of the constraints at $x^* = (1, 0)$:

$$\nabla g_1(x) = \big(3(x_1-1)^2,\ 1\big) \Rightarrow \nabla g_1(1,0) = (0,1), \qquad \nabla g_2(x) = (0,-1), \qquad \nabla g_3(x) = (1,0). \tag{1.45}$$

Equation (6). At x^*, $g_1(x^*) = 0$ and $g_2(x^*) = 0$, so both are active. The linearizing cone is

$$C_\ell^{(6)} = \{d : \ \nabla g_1(x^*)^\top d \le 0,\ \nabla g_2(x^*)^\top d \le 0\}. \tag{1.46}$$

These inequalities read

$$(0,1)\cdot(d_1,d_2) \le 0 \Rightarrow d_2 \le 0, \qquad (0,-1)\cdot(d_1,d_2) \le 0 \Rightarrow -d_2 \le 0 \Rightarrow d_2 \ge 0. \tag{1.47}$$

Thus $d_2 = 0$ and d_1 is free:

$$C_\ell^{(6)} = \{(d_1,0) : \ d_1 \in \mathbb{R}\}. \tag{1.48}$$

Since $C_{dd}(x^*) = \{d : \ d_1 > 0\}$, we obtain

$$C_{dd}(x^*) \cap C_\ell^{(6)} \neq \emptyset, \tag{1.49}$$

so condition (5) fails for problem (6), even though x^* is the minimizer.

Equation (7). Now $g_3(x^*) = 0$ is also active, and the linearizing cone becomes

$$C_\ell^{(7)} = \{d : \ \nabla g_1(x^*)^\top d \le 0,\ \nabla g_2(x^*)^\top d \le 0,\ \nabla g_3(x^*)^\top d \le 0\}. \tag{1.50}$$

The additional inequality is

$$(1,0)\cdot(d_1,d_2) \le 0 \Rightarrow d_1 \le 0. \tag{1.51}$$

Together with $d_2 = 0$ from the first two constraints, we get

$$C_\ell^{(7)} = \{(d_1,0) : \ d_1 \le 0\} = C_{fd}(x^*). \tag{1.52}$$

Therefore

$$C_{dd}(x^*) \cap C_\ell^{(7)} = \emptyset, \tag{1.53}$$

so condition (5) holds for problem (7).

(g-vii) Why this shows that (5) is not necessary in general You have:

- Condition (3) is always necessary for a local minimizer.
- Since $C_{fd}(x_0) \subseteq C_\ell(g,h,x_0)$, condition (5) implies (3).
- However, problem (6) has a (unique) minimizer x^* but (5) fails, so (5) is *not* necessary in general.

The reason is that C_ℓ can be a strict superset of C_{fd}: it contains directions that look feasible to first order, but are not truly feasible once higher-order terms matter.

1.2.10 Part (h) KKT conditions in simple words (including linear dependence)

Geometric meaning. At an unconstrained local minimizer, $\nabla f(x^*) = 0$. Under constraints, $\nabla f(x^*)$ need not be zero, but it must be balanced by the normals of the active constraints. KKT states that there exist multipliers λ_i and μ_j such that

$$\nabla f(x^*) + \sum_{i=1}^{m} \lambda_i \nabla g_i(x^*) + \sum_{j=1}^{p} \mu_j \nabla h_j(x^*) = 0, \tag{1.54}$$

with complementary slackness and sign conditions

$$\lambda_i \ge 0 \quad \forall i, \qquad \lambda_i g_i(x^*) = 0 \quad \forall i. \tag{1.55}$$

Thus only active inequalities may have $\lambda_i > 0$.

Why linear dependence matters. To guarantee existence of multipliers at a local minimizer, one typically needs a *constraint qualification* (e.g. LICQ), which informally says that the gradients of active constraints should not be "degenerate" (e.g. linearly dependent in a problematic way). If a qualification fails, a local minimizer may exist but KKT multipliers may not.

(i) Figure 3: four candidates, only one KKT point

How to reason. For a strictly convex quadratic-like f, the constrained minimizer is the point in the rectangle that first touches the lowest contour line. That point is typically a corner (unless the unconstrained minimizer lies inside the rectangle). Only at that corner do the active constraint normals generate a cone that can balance ∇f with nonnegative multipliers. The other corners fail because ∇f points in a direction that cannot be expressed as a nonnegative combination of the active normals there.

1.2.11 Part (j) Prove: KKT at x^* implies condition (5)

Show one implication of the equivalence: if KKT holds at x^*, then

$$C_{dd}(x^*) \cap C_\ell(g, h, x^*) = \emptyset. \tag{1.56}$$

Proof. Assume KKT holds at $x^* \in \mathcal{F}$:

$$\nabla f(x^*) + \sum_{i \in A(x^*)} \lambda_i \nabla g_i(x^*) + \sum_{j=1}^{p} \mu_j \nabla h_j(x^*) = 0, \qquad \lambda_i \geq 0. \tag{1.57}$$

Take any $d \in C_\ell(g, h, x^*)$. Then

$$\nabla g_i(x^*)^\top d \leq 0 \ \ \forall i \in A(x^*), \qquad \nabla h_j(x^*)^\top d = 0 \ \ \forall j. \tag{1.58}$$

Dot the KKT stationarity equation with d:

$$\nabla f(x^*)^\top d + \sum_{i \in A(x^*)} \lambda_i \nabla g_i(x^*)^\top d + \sum_{j=1}^{p} \mu_j \nabla h_j(x^*)^\top d = 0. \tag{1.59}$$

The last sum is zero because $\nabla h_j(x^*)^\top d = 0$. In the middle sum, each term satisfies $\lambda_i \geq 0$ and $\nabla g_i(x^*)^\top d \leq 0$, hence $\lambda_i \nabla g_i(x^*)^\top d \leq 0$. Therefore,

$$\nabla f(x^*)^\top d = -\sum_{i \in A(x^*)} \lambda_i \nabla g_i(x^*)^\top d \geq 0. \tag{1.60}$$

So no $d \in C_\ell(g, h, x^*)$ can satisfy $\nabla f(x^*)^\top d < 0$, which is exactly the definition of descent directions. Hence

$$C_{dd}(x^*) \cap C_\ell(g, h, x^*) = \emptyset. \tag{1.61}$$

1.2.12 Part (k) Check KKT for Equations (6) and (7), compute multipliers, and discuss uniqueness

Equation (6): constraints $g_1 \leq 0$, $g_2 \leq 0$

At $x^* = (1, 0)$, we have

$$\nabla f(x^*) = (-2, 0), \qquad \nabla g_1(x^*) = (0, 1), \qquad \nabla g_2(x^*) = (0, -1). \tag{1.62}$$

KKT stationarity would require multipliers $\lambda_1, \lambda_2 \geq 0$ such that

$$\nabla f(x^*) + \lambda_1 \nabla g_1(x^*) + \lambda_2 \nabla g_2(x^*) = (0, 0), \tag{1.63}$$

i.e.

$$(-2, 0) + \lambda_1(0, 1) + \lambda_2(0, -1) = (0, 0). \tag{1.64}$$

The first component yields $-2 = 0$, which is impossible. Therefore there are *no* KKT multipliers for (6), and KKT fails at the minimizer.

Equation (7): constraints $g_1 \le 0$, $g_2 \le 0$, $g_3 \le 0$ Now also $\nabla g_3(x^*) = (1, 0)$. KKT stationarity requires $\lambda_1, \lambda_2, \lambda_3 \ge 0$ such that

$$(-2, 0) + \lambda_1(0, 1) + \lambda_2(0, -1) + \lambda_3(1, 0) = (0, 0). \tag{1.65}$$

From the first component,

$$-2 + \lambda_3 = 0 \qquad \Rightarrow \qquad \lambda_3 = 2. \tag{1.66}$$

From the second component,

$$\lambda_1 - \lambda_2 = 0 \qquad \Rightarrow \qquad \lambda_1 = \lambda_2. \tag{1.67}$$

With $\lambda_1, \lambda_2 \ge 0$, this yields infinitely many solutions:

$$\lambda_3 = 2, \qquad \lambda_1 = \lambda_2 \ge 0. \tag{1.68}$$

Thus KKT holds for (7), λ_3 is unique, while λ_1, λ_2 are not unique (any equal nonnegative pair works).

Exercise sheet 9 for Numerical Optimization

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Exercise 1 (**Constrained optimization**, 1+2+2 points)

We will now deal with constrained optimization problems. Let $D \subset \mathbb{R}^n$ be open and as well the objective function

$$f : D \subset \mathbb{R}^n \to \mathbb{R}$$

as the constraints

$$g_1, \dots, g_m,\, h_1, \dots, h_p : D \to \mathbb{R}$$

be continuously differentiable. We define corresponding index sets

$$\mathcal{I} := \{1, \dots, m\} \quad \text{and} \quad \mathcal{E} := \{1, \dots, p\}.$$

The constrained optimization problem reads

$$\min_{x \in D} f(x) \quad \text{such that} \quad g_i(x) \leq 0 \ \forall i \in \mathcal{I} \quad \text{and} \quad h_j(x) = 0 \ \forall j \in \mathcal{E} \ . \tag{1}$$

So we try to minimize f under the constraint that some equalities and inequalities are fulfilled. It can also be just equalities or inequalities, so $\mathcal{I}$ and $\mathcal{E}$ are allowed to be empty. If both are empty, we end up at unconstrained optimization.
The set of all points satisfying the constraints is called *feasible set*,

$$\mathcal{F} := \{x \in D \,|\, g_i(x) \leq 0 \ \forall i \in \mathcal{I},\ h_j(x) = 0 \ \forall j \in \mathcal{E}\} \ .$$

With that, problem (1) becomes

$$\min_{x \in \mathcal{F}} f(x) \ .$$

Consider the exemplary constrained optimization problem

$$\min_{x \in \mathbb{R}^2} x_1 + x_2 \quad \text{such that} \quad x_1^2 \leq 1 - x_2^2 \quad \text{and} \quad x_1 = x_2 \ . \tag{2}$$

a) Indicate the objective function and the constraint functions such that (2) takes the form of (1). Also give the corresponding index sets.

b) Draw the subset of $\mathbb{R}^2$ that fulfils the first constraint, the one that fulfils the second constraint and the feasible set $\mathcal{F}$.

c) There is a unique minimizer of the exemplary problem (2). What is this unique minimizer and what is the corresponding objective function value?

Exercise 2 (**Derivation of necessary optimality conditions (KKT conditions)**, 3+3+4+3+3+4+1+1+2+2+2+2+2+3+3+4+3 points)

Our goal is to find computable optimality conditions for the constrained optimization problem (1). For that, we first have to clarify how one can move inside the feasible set. For unconstrained optimization, we could move everywhere inside the domain of f. But now, the possibilities to move are restricted.

Definition 1 (Feasible direction) Suppose a point $x_0 \in \mathcal{F}$. A direction $d \in \mathbb{R}^n$ is called *feasible* if

$$\exists \delta > 0 \quad \text{such that} \quad x_0 + td \in \mathcal{F} \quad \forall t \in [0, \delta] \ ,$$

meaning that one can move along d for a certain distance without leaving $\mathcal{F}$.

We are interested in the set of all feasible directions $C_{fd}(x_0)$.

a) Show that $C_{fd}(x_0)$ is a cone.

What is the connection between the feasibility of a direction and the constraints?

- The index set

  $$\mathcal{A}(x_0) := \{i \in \mathcal{I} \,|\, g_i(x_0) = 0\}$$

  describes those inequality constraints that become equalities at x_0. They are called *active*. These constraints are critical as there might be directions $d \in \mathbb{R}^n$ with $g_i(x_0 + td) > 0$ $\forall t > 0$.
- The *inactive* inequality constraints, indicated by the index set

  $$\mathcal{I} \setminus \mathcal{A}(x_0) = \{i \in \mathcal{I} \,|\, g_i(x_0) < 0\} \ ,$$

  are not important as due to continuity for every direction $d \in \mathbb{R}^n$ there exists $s > 0$ such that $g_i(x_0 + sd) < 0$.
- The equality constraints are critical as there might be directions $d \in \mathbb{R}^n$ with $h_j(x_0+td) \neq 0$ $\forall t > 0$.

Consequently, only the active inequality and the equality constraints are relevant for the feasibility of directions.

b) Consider the constrained optimization problem (2) from exercise 1. Is the inequality constraint active at the point $p_0 = (0, 0)$? And at $p_1 = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$? Based on your drawing of the feasible set from exercise 1b), determine $C_{fd}(p_0)$ and $C_{fd}(p_1)$. Make yourself clear what influence the inequality constraint has on the set of feasible directions, depending on whether it is active or not.

Definition 2 (Cone of the descent directions) The *cone of the descent directions* at x_0 is

$$C_{dd}(x_0) := \left\{d \in \mathbb{R}^n \,|\, \nabla f(x_0)^T d < 0\right\} \ .$$

Note: On exercise sheet 8 we defined a cone as a set K with $\alpha x \in K$ for every $x \in K$ and $\alpha \geq 0$. According to this definition, every non-empty cone must contain the zero vector $\mathbf{0}$. But $C_{dd}(x_0)$ cannot contain the zero vector and therefore cannot be a cone. If one slightly changes the definition of a cone to a set K with $\alpha x \in K$ for every $x \in K$ and $\alpha > 0$, $C_{dd}(x_0)$ is a cone. That is how cones are defined in the lecture notes. And therefore I will also call $C_{dd}(x_0)$ a cone.

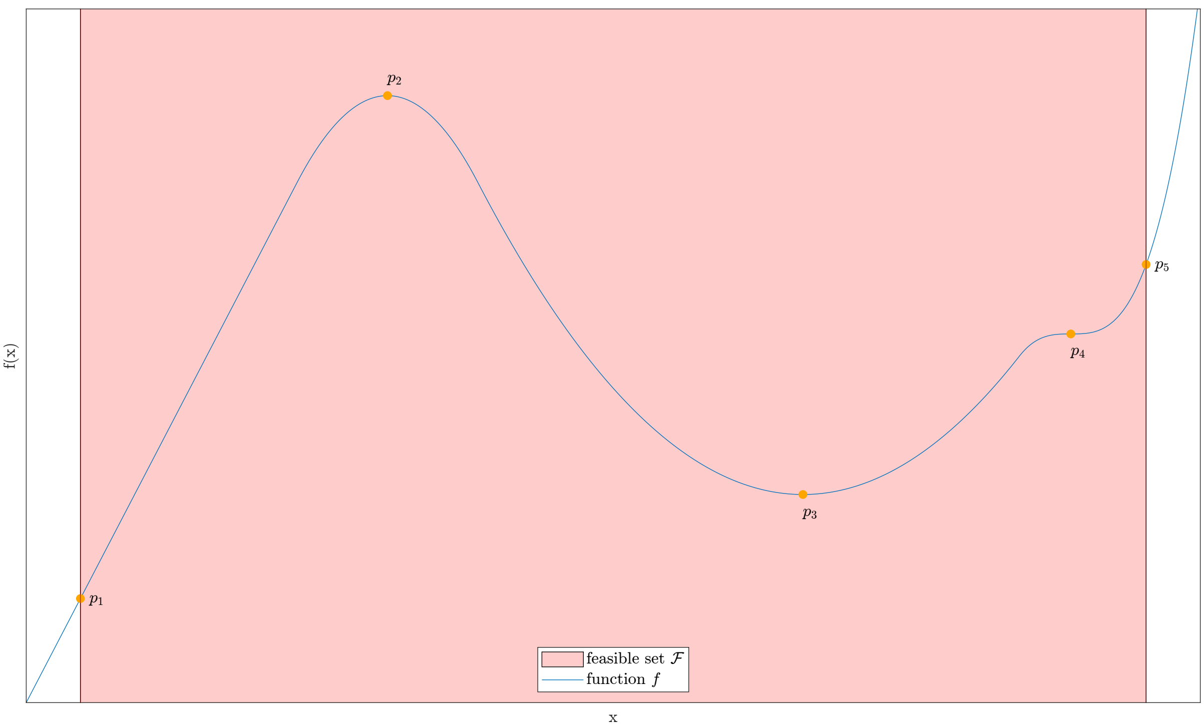

Figure 1: A function $f : \mathbb{R} \to \mathbb{R}$ and a feasible set $\mathcal{F} \subset \mathbb{R}$.

c) Prove that if $x^\star$ is a local minimizer of (1), then

$$C_{dd}(x^\star) \cap C_{fd}(x^\star) = \emptyset \ . \tag{3}$$

d) What does condition (3) mean in simple words? To do this, consider the orange marked points in the plot in figure 1. Why is this condition necessary for a local minimizer? And why not sufficient?

Summarized, condition (3) is a necessary condition for a local minimizer of the constrained optimization problem, but not a sufficient condition.
One might think we have achieved our aim, as we have a necessary condition for a local minimizer. Unfortunately, in general it is not possible to compute the set of feasible directions $C_{fd}(x_0)$. And an optimality condition that cannot be determined does not get us anywhere.

To proceed, we consider a superset of the set of feasible directions, the *linearizing cone* $C_l(g, h, x_0)$.

Definition 3 (Linearizing cone) For $x_0 \in \mathcal{F}$, the *linearizing cone* is defined as

$$C_l(g, h, x_0) := \left\{ d \in \mathbb{R}^n \,|\, \nabla g_i(x_0)^T d \leq 0 \ \forall i \in \mathcal{A}(x_0), \ \nabla h_j(x_0)^T d = 0 \ \forall j \in \mathcal{E} \right\} \ .$$

e) What is the linearizing cone in simple words? To do this, consider the points in the plots in figure 2. What is the linearizing cone in each case? Why must every feasible direction lie in the linearizing cone?

f) Show that for $x_0 \in \mathcal{F}$ it is

$$C_{fd}(x_0) \subset C_l(g, h, x_0) \ . \tag{4}$$

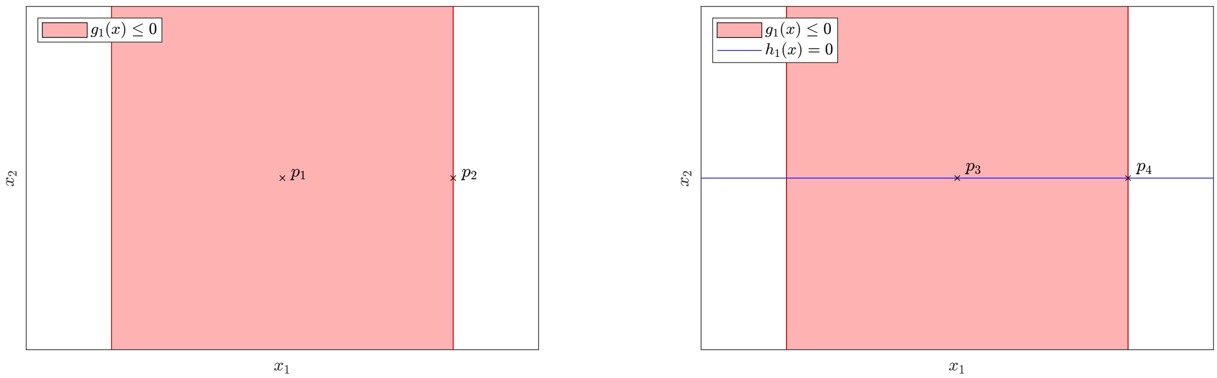

Figure 2: Two feasible sets $\mathcal{F} \subset \mathbb{R}^2$.

Due to (4), the condition

$$C_{dd}(x^\star) \cap C_l(g, h, x^\star) = \emptyset \tag{5}$$

implies (3). But that does not mean that condition (5) is necessary for a local minimizer.

g) With

$$f(x) = (x_1 - 2)^2 + x_2^2\,, \quad g_1(x) = (x_1 - 1)^3 + x_2\,, \quad g_2(x) = -x_2 \quad \text{and} \quad g_3(x) = x_1 - 1$$

consider the constrained problems

$$\min_{x \in \mathbb{R}^2} f(x) \quad \text{such that} \quad g_1(x) \leq 0 \quad \text{and} \quad g_2(x) \leq 0 \tag{6}$$

and

$$\min_{x \in \mathbb{R}^2} f(x) \quad \text{such that} \quad g_1(x) \leq 0\,, \quad g_2(x) \leq 0 \quad \text{and} \quad g_3(x) \leq 0\,. \tag{7}$$

i) Draw the feasible sets of both problems.
ii) Why do both problems have a unique minimizer and what are the respective minimizers?
iii) At the respective minimizers, what are the cones of descent directions?
iv) At the respective minimizers, what are the cones of feasible directions?
v) Draw the cones of descent directions and the cones of feasible directions at the respective minimizers. Is condition (3) fulfilled at the respective minimizers and why does it have to be?
vi) At the respective minimizers, what are the linearizing cones? Draw them.
vii) Is condition (5) fulfilled at the respective minimizers? What is the reason for that condition (5) is fulfilled for one problem but not for the other, and what are no reasons? What does that mean for condition (5) being necessary for a local minimizer?

Condition (5)'s right to exist is that it is equivalent to a condition that is easy to compute, the so-called *KKT conditions*.

Definition 4 (KKT conditions) The *KKT conditions* assume that for $x_0 \in \mathcal{F}$ there exist so-called *Lagrange multipliers* $\lambda \in \mathbb{R}^m_{\geq 0}$ and $\mu \in \mathbb{R}^p$ such that

$$\nabla f(x_0) + \sum_{i=1}^{m} \lambda_i \nabla g_i(x_0) + \sum_{j=1}^{p} \mu_j \nabla h_j(x_0) = \mathbf{0}$$

and

$$\lambda_i g_i(x_0) = 0 \;\forall i \in \mathcal{I}\,.$$

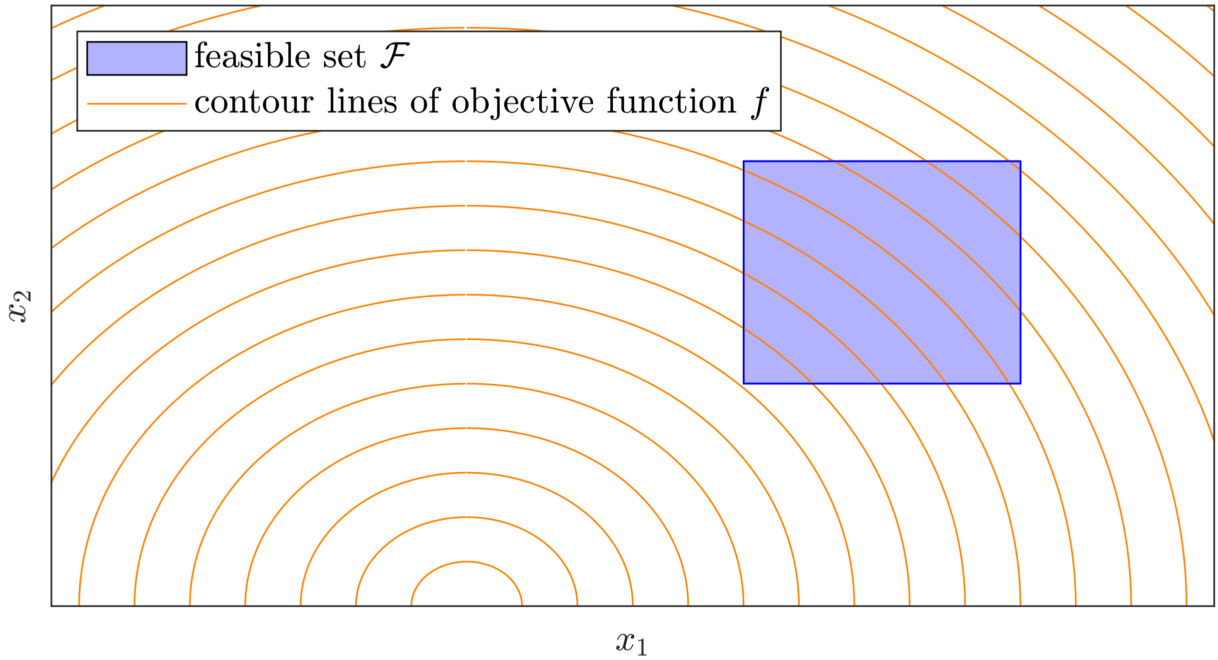

Figure 3: Sketch of an optimization problem in $\mathbb{R}^2$. The objective function f is supposed to be an upward-opened parabola.

h) What do the KKT conditions say in simple words? Especially about linear dependence.

i) Consider figure 3. There are four points in the feasible set, that are candidates for being a KKT point. Which are they and why?
Actually, there is only one KKT point. Which is it and why?

The KKT conditions being fulfilled is equivalent to condition (5).

j) For simplicity's sake, we only want to prove one implication of the equivalence. Prove that if the KKT conditions are fulfilled at $x^\star \in \mathcal{F}$, then condition (5) is satisfied.

k) For both example problems from exercise g), check if the KKT conditions are satisfied at the respective unique minimizers. If so, compute the Lagrange multipliers. Are they unique? Verify that your results comply with condition (5).

That's it for this sheet. We have the implications marked in black in the following logical chain:

$$\begin{array}{rcl} x^\star \text{ local minimizer} & \Longrightarrow & C_{dd}(x^\star) \cap C_{fd}(x^\star) = \emptyset \\ & & \Uparrow \ \Downarrow \\ & & C_{dd}(x^\star) \cap C_l(g,h,x^\star) = \emptyset \iff \text{KKT holds} \end{array}$$

With that, the KKT conditions are not necessary for a local minimizer yet. On the next sheet, we will take care of the implication marked in red and obtain necessity. However, the implication marked in red will not apply to every optimization problem, but only to those that satisfy so-called *constraint qualifications*. You have already seen the consequences of this in exercise g), where one of two seemingly identical optimization problems was formulated unfavourably and therefore did not satisfy the KKT conditions at a local minimizer.